PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Association of frailty and cognitive function disorders in old patients with COVID-19: a protocol of systematic review and meta-analysis
AUTHORS	jiao, bo; Chen, Mingyuan; Fan, Minmin; Luo, Xi; Chen, Chan; Liu, Jin

VERSION 1 – REVIEW

REVIEWER	Sara Salini Fondazione Policlinico Universitario Agostino Gemelli IRCCS
REVIEW RETURNED	08-Nov-2021

GENERAL COMMENTS	Dear Author, congratulations on your choice of topic. I think it is really important to investigate this aspect in Covid-19 patients. Personally, I have been dealing with frailty for a short time and I think the topic is very relevant in determining specific outcomes. Together with the group I work with, we have recently published a paper regarding frailty and Covid-19, with interesting results. I hope your results provide relevant insights into the management of frailty patients with Covid-19. Thank you very much and good luck with your work.

REVIEWER	Fabrizia Lattanzio Italian National Research Center on Aging, Scientific Direction
REVIEW RETURNED	12-Nov-2021

GENERAL COMMENTS	The aim of the present protocol is of a great interest and the abstract introduces adequately the methodology the authors intend to apply. Unfortunately, the English language makes very difficult to understand or to interpret the meaning of whole phrases. I would suggest an extensive revision. From a methodological point of view, I would like to suggest the following advices: 1) "Participants": please specify the age limit for adults (18+). Please take into consideration the opportunity of splitting results according to specific age groups (eg. separating older adults 65+ or 75+ from the others) 2) When planning to perform risk of bias assessment and quality, please take into account the opportunity of reporting also the following measures: Inconsistency, Indirectness, Imprecision, Sensitivity, Specificity and Pooled C-statistic (as for the NICE guidelines. FYI see: https://www.ncbi.nlm.nih.gov/books/NBK409218/#ch7.s36) 3) Authors need to review the Cochrane Handbook for handling non-randomised studies in a systematic review. Caution is needed before pooling studies 4) The authors need to be more explicit about the primary outcome,

	i.e. how do they plan to assess cognitive disorder?

VERSION 1 – AUTHOR RESPONSE

Reviewer 1

Comment 1. *Dear Author, congratulations on your choice of topic. I think it is really important to investigate this aspect in Covid-19 patients. Personally, I have been dealing with frailty for a short time and I think the topic is very relevant in determining specific outcomes. Together with the group I work with, we have recently published a paper regarding frailty and Covid-19, with interesting results. I hope your results provide relevant insights into the management of frailty patients with Covid-19. Thank you very much and good luck with your work.*

Response: We are grateful for this reviewer's comment. We have read the article, named "Frailty Assessment in the Emergency Department for Risk Stratification of COVID-19 Patients Aged ≥80 Years", which was published by the reviewer and his group. They investigated the role of frailty in predicting the clinical risk of in-hospital deaths in COVID-19 patients aged ≥80 years at the time of emergency department presentation. We thank the reviewer and his group for their efforts in this research. And we will cite this article in our systematic review in the future.

Reviewer 2

Comment 1. *The aim of the present protocol is of a great interest and the abstract introduces adequately the methodology the authors intend to apply.*
Unfortunately, the English language makes very difficult to understand or to interpret the meaning of whole phrases. I would suggest an extensive revision.

Response: We thank the reviewer for this comment. Our revised manuscript has been edited by a native English speaker with expertise in this field. The editing certificate has been provided in the supplementary file.

Comment 2. *"Participants": please specify the age limit for adults (18+). Please take into consideration the opportunity of splitting results according to specific age groups (eg. separating older adults 65+ or 75+ from the others)*

Response: We apologize for the confusing expression. We have corrected this, and now specified the age limit for the participants. Please see page7 line26, which reads as follows: “Elderly COVID-19 patients (≥65 years old), with cognitive impairment, will be considered”. In future, we will perform subgroup analysis based on the age group in the paper of formal systematic review according the reviewer’s suggestion.

Comment 3. *When planning to perform risk of bias assessment and quality, please take into account the opportunity of reporting also the following measures: Inconsistency, Indirectness, Imprecision, Sensitivity, Specificity and Pooled C-statistic (as for the NICE guidelines. FYI see: https://www.ncbi.nlm.nih.gov/books/NBK409218/#ch7.s36)*

Response: We thank the reviewer for this comment. We have included the NICE guidelines to our paper for assessing the risk of bias and quality. The aforementioned evaluation measures (Inconsistency, Indirectness, Imprecision, Sensitivity, Specificity and Pooled C-statistic) will be fully considered during quality assessment in the meta-analysis.

Comment 4: *Authors need to review the Cochrane Handbook for handling non-randomised studies in a systematic review. Caution is needed before pooling studies.*

Response: We are grateful for this reviewer’s comment. We will review the Cochrane Handbook again. In the formal systematic review, we will pool the included studies based on this guideline. This will be done cautiously.

Comment 5: *The authors need to be more explicit about the primary outcome, i.e. how do they plan to assess cognitive disorder?*

Response: We apologize for the confusing expression. We have now introduced, in the revised manuscript, a tool called the Brief Assessment of Impaired Cognition (BASIC), which focuses on the Self-report, Informant report, Supermarket Fluency, and Category Cued Memory Test, to assess cognitive disorders.